

DL for ENG

# Who Is Teaching This

REMUS TEODORESCU

- **Professor, AAU Energy** — power electronics, batteries, and control for the grid
- **thirty years of physics-based models** — and five of asking what data adds to them
- **this course** — twelve lectures written, and rewritten, over the past year
- **with Noman Khan** — research assistant, who runs the exercise sessions with you and has reviewed every notebook
- **the honest part** — the material was developed with AI assistance. I will say so wherever it matters.

## WHY THIS COURSE EXISTS

|                  |                                      |
|------------------|--------------------------------------|
| the problem      | energy systems have physics AND data |
| the usual answer | pick one and ignore the other        |
| this course      | put the physics inside the learning  |
| the name for it  | physics-informed deep learning       |

## A FIRST EDITION

Everything here is new this year. You are its first readers, and its reviewers.

# The Course in One Picture

## ONE METHOD, TWELVE LECTURES

- **Part 1 · L1 – L6** — what a network is, how it learns, and why it works
- **the join · L6.1** — a loss term need contain no data at all
- **Part 2 · L7 – L12** — the physics goes into the loss
- **the applications** — heat, fluid flow, batteries and fuel cells, a robot car, the power grid
- **L13** — your mini-project, on your own topic

## WHERE YOUR PROJECT MIGHT LAND

|                        |                                          |
|------------------------|------------------------------------------|
| <b>thermal</b>         | L8 — heat transfer, steady and transient |
| <b>flow</b>            | L9 — Burgers, channel flow               |
| <b>electrochemical</b> | L10 — Li-ion cells, solid oxide cells    |
| <b>autonomous</b>      | L11 — camera navigation on a JetRacer    |
| <b>grid</b>            | L12 — state estimation, stability        |

*If your project is none of these, that is not a problem. It is the sixth row, and mini-projects are chosen at L13.*

# Who Is in the Room

## THIRTY SECONDS EACH

- **your name** — and the programme you are on
- **your background** — mechanical, electrical, energy, something else
- **your semester project** — in one sentence
- **what it is trying to do** — predict, control, design, or understand
- **your Part 2 topic** — heat, flow, electrochemistry, robots, or the grid: the one closest to your programme or project. Decided today; it fixes which Part 2 exercise you hand in.

## SEVEN PROGRAMMES, FIVE PART 2 TOPICS

**PED • EPSH** power grid stability • Ex12

**MCE** robot navigation • Ex11

**HYTEC** electrochemical devices • Ex10

**OES** fluid flow • Ex09

**APEL • AIAS** heat transfer • Ex08, or by project

*A default by programme; your semester project can override it. The exercises cross all five, so different backgrounds are an asset here.*

## THE QUESTION FOR THE ROOM

In one sentence: what is your semester project trying to predict, control, or design — and which of the five Part 2 topics is closest to it?

# Eight Weeks, Thirteen Lectures

## PART 1

|         |                                                         |
|---------|---------------------------------------------------------|
| L1 · L2 | 2 – 9 Sep · at home · recorded — Python, NumPy, PyTorch |
| L3      | Wed 9 Sep · 12:30 – 16:15 · today                       |
| L4      | Wed 16 Sep · 12:30 – 16:15                              |
| L5      | Mon 21 Sep · 12:30 – 16:15                              |
| L6      | Fri 25 Sep · 10:15 – 15:15                              |

## PART 2

|     |                                           |
|-----|-------------------------------------------|
| L7  | Wed 30 Sep · 12:30 – 16:15                |
| L8  | Fri 2 Oct · 10:15 – 14:15                 |
| L9  | Wed 7 Oct · 12:30 – 15:15                 |
| L10 | Wed 14 Oct · 12:30 – 16:15                |
| L11 | Fri 16 Oct · 10:15 – 14:15 · ESBJERG      |
| L12 | Mon 26 Oct · 12:30 – 16:15                |
| L13 | Thu 29 Oct · 8:15 – 12:00 · mini-projects |

*All sessions in Aalborg except L11, the JetRacer day in Esbjerg. Each lecture is two sublectures with a break; L3.2 is recorded, like L1 and L2.*

# Live Lectures and Recorded Ones

## TWO TRACKS, SAME MATERIAL

- **recorded only** — L1, L2, and L3.2: watch at your own pace, before the lecture that needs them
- **live** — everything else, in this room, with a recorded version published afterwards for review
- **pause points** — each recording stops twice and asks you to work something out before it continues
- **the recordings are narrated by a synthetic voice from my script** — say if that gets in the way

## HOW TO USE A RECORDING

|               |                                        |
|---------------|----------------------------------------|
| <b>before</b> | watch, and stop at the pause points    |
| <b>during</b> | have the notebook open beside it       |
| <b>after</b>  | replay the ten questions at the end    |
| <b>do not</b> | watch at double speed the night before |

## THE QUESTION FOR THE ROOM

Did you watch L1 and L2 — all, some, skimmed, or not yet? And what would have made you watch more?

# The Exercises Run in Your Browser

## GOOGLE COLAB, NOTHING TO INSTALL

- [github.com/RemusTeodorescu/DL-for-Engineers-Public-Course](https://github.com/RemusTeodorescu/DL-for-Engineers-Public-Course) — every lecture PDF and every exercise notebook, always the latest version
- **click the badge** — Open in Colab, at the top of each notebook
- **run the first cell** — it fetches the set's library files by itself
- **first, save a copy in Drive** — or your work vanishes when the tab closes
- **a Google account is all you need** — no Python installed, no GPU of your own

### THE ONE HABIT THAT SAVES EVENINGS

|                  |                                             |
|------------------|---------------------------------------------|
| <b>File</b>      | Save a copy in Drive — before anything else |
| <b>then</b>      | every edit auto-saves                       |
| <b>the badge</b> | always reopens the pristine original        |
| <b>lost work</b> | almost always an unsaved tab                |

### THE QUESTION FOR THE ROOM

Who has opened a notebook in Colab already? Did the first cell run without asking for anything?

# Exercise, Report, Moodle — Within a Week

## THE RHYTHM

- **each lecture has an exercise set** — notebooks in numbered order, ending in a report with your measured numbers
- **Part 1, Ex01 - Ex06** — mandatory for everyone: six reports, one per set
- **Part 2, one set mandatory** — the topic from the intro round, your seventh report; the other sets are open, not handed in
- **hand in on Moodle within one week** — the reports are the course work, and the oral exam builds on them
- **late** — say so on Moodle before the deadline, and we agree a date

## YOUR SEVEN REPORTS

|                      |                                                                                  |
|----------------------|----------------------------------------------------------------------------------|
| Ex01 - Ex06          | everyone · Part 1 · six reports                                                  |
| + one of Ex08 - Ex12 | heat · flow · electrochemical · robots · grid,<br>the topic from the intro round |
| numbers              | yours, not the slide's                                                           |
| a judgement          | which model, and why — defended                                                  |
| a check              | what you tried that did not work                                                 |
| the seeds            | if a number depends on chance, say so                                            |

## THE QUESTION FOR THE ROOM

Ex01 and Ex02 — how far did you get, and which cell cost the most time: the Python, the maths, or the notebook environment?

# The Mini-Project

## YOUR TOPIC, THIS METHOD

- **chosen at L13** — 29 October, after all twelve lectures
- **close to your semester project** — the point is to use this on something you already care about
- **alone or in a group** — your choice; a group shares the modelling, not the writing
- **the shape** — a physics-informed model of one thing from your project, with a measured comparison
- **support** — Noman in every exercise session, the L13 session, and the Moodle forum until the hand-in

## WHAT COUNTS AS A RESULT

|                       |                                          |
|-----------------------|------------------------------------------|
| <b>a baseline</b>     | the physics-only or data-only answer     |
| <b>the model</b>      | physics in the loss, trained by you      |
| <b>the comparison</b> | measured, with its uncertainty           |
| <b>the judgement</b>  | would you use it? Under what conditions? |

## THE QUESTION FOR THE ROOM

Looking at the board from the intro round: which of your projects could become a mini-project, and which two might share one?

# Where Things Happen

## ONE PLACE FOR EVERYTHING

- **Moodle** — announcements, hand-ins, the forum, links to the videos
- **the public repository** — slides and notebooks, always the latest version
- **the Moodle forum before email** — a question one of you has, ten of you have; Noman and I both read it
- **I read the forum before each lecture** — and answer what needs answering there, for everyone at once
- **exercise sessions** — Noman Khan is there for every one; office hours with me by appointment through Moodle

## THE FEEDBACK DEAL

|                 |                                                                 |
|-----------------|-----------------------------------------------------------------|
| <b>you find</b> | a cell that fails, a claim that is wrong, a slide that confuses |
| <b>you post</b> | on the Moodle forum, with the notebook and cell                 |
| <b>I fix</b>    | usually the same week, and say so                               |
| <b>you see</b>  | the fix on the badge, immediately                               |

## THE QUESTION FOR THE ROOM

What is one thing you would like different by next week — in the videos, the notebooks, or the way the sessions run?

# You Will Use a Language Model. Everyone Does.

## THE ONLY QUESTION IS HOW

- **paste a TODO cell into any model** — it will write the answer in seconds, and it will usually be right
- **so the assignment is not the point** — the assignment is the occasion
- **what I assess** — whether you can explain, defend, and adapt what you hand in
- **what an engineer is paid for** — the judgement, not the typing
- **the rest of this part** — what the difference looks like, and the rule for this course

## TWO STUDENTS, SAME SUBMISSION

|             |                                                    |
|-------------|----------------------------------------------------|
| student A   | pasted, ran, submitted. Twenty minutes.            |
| student B   | asked, read, changed a line, asked why. Two hours. |
| in November | identical reports                                  |
| in March    | only one can build the next thing                  |

## THE QUESTION FOR THE ROOM

Hands up: who used a language model on Ex01 or Ex02? Nobody in this room will be marked down for that answer.

# From General Questions to Your Engineering Problem

## YOU ALREADY USE THESE TOOLS — JUST NOT FOR THIS

- **how most people use them** — summarise, translate, draft, explain a concept, fix a line of code
- **what an engineering problem needs** — a specific system, units, an equation, a measurement to check against, a model somebody must trust
- **the gap** — a general answer is not an engineering answer. It has no plant, no sensor, no failure mode.
- **what this course teaches** — how to take a tool that answers general questions into your own specific project, step by step
- **the progression** — a Python tutor in L1 – L2, a reasoning partner in Part 1, a collaborator on your physics in Part 2, a colleague on your mini-project

## THE SHIFT, IN FOUR MOVES

|                  |                                         |
|------------------|-----------------------------------------|
| from a topic     | to your system, with its numbers        |
| from an answer   | to a prediction you then measure        |
| from trusting it | to checking it against the physics      |
| from asking      | to specifying, and defending the result |

## THE QUESTION FOR THE ROOM

What was the last thing you asked a language model — and was it anything like an engineering problem?

# The Rule for This Course

## ALLOWED, FOR EVERYTHING

Any tool, any time: writing code, debugging, explaining a slide, drafting a report. This is how engineering is done now, and pretending otherwise would teach you a lie.

## DECLARED, IN THE REPORT

One line: which tool, for what. Not a confession — a method statement, like naming the solver you used.

## OWNED, LINE BY LINE

Anything you hand in, you can explain when asked. If you cannot, it is not yours yet — go back and make it yours.

## REPORTS, EXAM, MINI-PROJECT

|                       |                                                    |
|-----------------------|----------------------------------------------------|
| written reports       | always state which tool you used, and for what     |
| exam · preparation    | everything allowed, any tool                       |
| exam · the 20 minutes | nothing allowed — you, the board, and the question |
| in the mini-project   | a live discussion of your model                    |
| never                 | a tool detector, or a trap                         |

*The twenty minutes are the whole point of the rule above: what you own, you can explain without a tool.*

# What Deeper Learning Looks Like

## THREE THINGS A PASTE CANNOT GIVE YOU

- **prediction** — say what a change will do before you run it. Right or wrong, you learn either way.
- **failure** — the exercises include runs that go wrong on purpose. The report asks what you saw.
- **transfer** — the same idea in a new place. Weight sharing appears three times in two lectures; noticing is the learning.
- **the model can do all three with you** — ask it to quiz you, not to answer you

## PROMPTS THAT TEACH

|                     |                                       |
|---------------------|---------------------------------------|
| "before I run this" | "what should the loss do, and why?"   |
| "I got X"           | "what are three reasons it is not Y?" |
| "explain the line"  | "then ask me to explain it back"      |
| "disagree with me"  | "where is my reasoning weakest?"      |

## THE QUESTION FOR THE ROOM

Which of those four prompts have you ever used — and which would feel strange to try?

# Where the Energy Sector Is Hiring

## THE INTERSECTION

- **most AI courses teach you to fit data** — most energy courses teach you the physics
- **this one sits at the intersection** — and that is where the sector is hiring
- **in engineering you usually know the equation** — discard it and fit a curve, and the model fails the moment conditions leave the training data
- **physics in the loss instead** — the model respects conservation, and can be trusted further from the data it saw
- **not a future skill** — the companies on the right are doing it now

## IN PRODUCTION, IN DENMARK AND BEYOND

### Energinet

AI forecast of reserve capacity on the Danish grid — about a million kroner a week saved

### Ørsted

machine learning across 5.5 GW of wind, solar and storage

### Vestas · Grundfos

fleet-wide blade data; machine-health models against unplanned downtime

### Siemens Energy · NVIDIA

an industrial AI operating system across the life of physical plant

## THE QUESTION FOR THE ROOM

You will graduate into an industry that expects physics and networks together. Which of these companies, or which other, do you have your eye on?

# Four Things You Will Be Able To Do

## BUILD A MODEL FROM DATA, AND SAY WHAT IT ASSUMES

A trained network, and the sentence about its noise model, its capacity, and the seeds it was checked on. Most published results skip that sentence.

## PUT AN EQUATION INSIDE THE LEARNING

A physics-informed network for a heat, flow, cell, or grid problem — trained with no measurements at all, then with some.

## DEPLOY ONE, AND MEASURE WHAT IT COSTS

Quantised, timed, on a device that also drives the motors. Latency budget in metres, not milliseconds.

## AND THE FOURTH

|         |                                            |
|---------|--------------------------------------------|
| judge   | when not to use any of it                  |
| because | a solver that runs in a second wins        |
| because | thin data deserves a curve, not a network  |
| because | certification wants bounds, not error bars |

*The fourth is the one that makes the other three worth having. It is also the one interviewers ask about.*

# Back to Your Projects

## THE LIST FROM THE INTRO ROUND

- **for each project on the board** — which of the four capabilities would move it forward?
- **the honest answer for some** — none yet, or not this semester. That is fine.
- **the mini-project is where this closes** — your topic, this method, measured
- **and the semester project is the reason to care** — not the grade

## WHAT TO DO THIS WEEK

|                                 |                                          |
|---------------------------------|------------------------------------------|
| <b>today</b>                    | L3.1 after the break; L3.2 on video      |
| <b>this week</b>                | Ex03 — one notebook, your first model    |
| <b>by Wed 16 Sep</b>            | the Ex03 report on Moodle                |
| <b>if L1 – L2 are unwatched</b> | watch them before Ex03. It assumes them. |

## THE QUESTION FOR THE ROOM

Look at your own project on the board: which one of the four capabilities is closest to it?

BREAK

# Fifteen Minutes. Then Lecture 3.1

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What artificial intelligence is, how it got here, and where an engineer should and should not reach for it. No mathematics beyond one energy function.

## WHAT I TAKE FROM THIS SESSION

|               |                                                |
|---------------|------------------------------------------------|
| the board     | your projects, under the five application rows |
| the videos    | what stopped you watching, if anything         |
| the notebooks | which cell cost the most time                  |
| one thing     | you want different by next week                |

*All four go into the material this week. You will see the changes on the badge, and I will say what changed at the start of L4.*

# If Asked: What Ex03 Contains

## YOUR FIRST MODEL, ONE NOTEBOOK

- a **vibrating mass** — a signal you can write the physics of, and measure
- **fit a network to it** — and watch it fail to extrapolate
- **then add the physics** — and watch the same network extrapolate
- **the report** — the extrapolation plot, and one paragraph on why
- **the point of the whole course** — in one notebook, before the theory

### IT ASSUMES

|             |                                        |
|-------------|----------------------------------------|
| from L1     | functions, arrays, shapes              |
| from L2     | a tensor, and <code>.backward()</code> |
| from nobody | any machine learning                   |
| time        | two to three hours                     |

# If Asked: How the Recordings Were Made

## HONESTLY

- **the slides** — generated from code, one script per lecture, so every deck agrees on style
- **the script** — written by me for a listener, with two pause points per lecture
- **the voice** — synthetic, from that script. It is not me. It says what I would say.
- **the cursor** — highlights the part of the slide the script is pointing at
- **the reason** — twelve lectures, each in a live and a recorded form, in one year

## WHAT AI DID, AND DID NOT

|                |                                           |
|----------------|-------------------------------------------|
| <b>did</b>     | draft, check, rewrite, render, narrate    |
| <b>did not</b> | decide what the course teaches            |
| <b>did not</b> | grade anything, ever                      |
| <b>will</b>    | get better when you tell me what is wrong |

# If Asked: The JetRacer Day

FRIDAY 16 OCTOBER, ESBJERG

- **six Waveshare JetRacer Pro cars** — a camera, a Jetson Nano, four 18650 cells
- **you drive, it records** — then a network learns to steer from your driving
- **then it drives** — and you measure how far it travels blind between two decisions
- **groups of three or four** — one car each, all morning
- **travel** — plan the trip to Esbjerg; details on Moodle by L9

## BEFORE THAT DAY

|                  |                                            |
|------------------|--------------------------------------------|
| L5.1             | convolutional networks — what the car sees |
| L6.2             | deployment — why the car cannot wait       |
| Ex11 notebook 00 | run it the week before                     |
| a charged laptop | and closed-toe shoes                       |

# If Asked: Reading

## TWO BOOKS, BOTH FREE ONLINE

- **Prince** — Understanding Deep Learning, MIT Press 2023. The main text for Part 1.
- **Goodfellow, Bengio, Courville** — Deep Learning, MIT Press 2016. For the optimisation chapters.
- **each lecture names its chapter** — usually twenty pages, read before or after
- **Part 2 has no textbook** — the papers are cited on the slides, and the exercises are the text
- **nothing to buy** — both books are free on their authors' sites

## FOR THIS WEEK

|                    |                                   |
|--------------------|-----------------------------------|
| <b>read</b>        | Prince chapter 1 — twenty minutes |
| <b>watch</b>       | L3.2 — what the field has built   |
| <b>run</b>         | Ex03                              |
| <b>bring to L4</b> | one model from your own work      |